

Big Picture Science and Modeling

What's the point of statistics?

- a process of learning through empirical observations

. . .

- together, science philosophy and statistical modeling is the backbone to empirical learning

Today

1. Study Objectives, Hypotheses, and Predictions
2. Big Data and Sampling
3. Inference and Prediction
4. Model-Based vs Design-Based Sampling/Inference

. . .

Lab: Simulation and Markdown

Study Objective

What you want to accomplish; can have multiple related objectives in a single manuscript.

Our objective is to understand the space-use of urban living coyotes.



Framing the importance of the objective(s) provides the *justification* and depends on the audience.

Hypothesis

1. A story that explains how the world works
2. An explanation for an observed phenomenon

Coyotes have small home ranges in urban areas

Research/Scientific Hypothesis

“A statement about a phenomenon that also includes the potential mechanism or cause of that phenomenon”. (Betts et al. 2021)

Coyotes have small home ranges in urban areas because resource density is high, leading to reduced ranging.

Statistical Model/Hypothesis

- An explicit mathematical and stochastic representation of the observational & mechanistic process of the empirical observations.

Statistical Model/Hypothesis

Example:

$$\mathbf{y} = \beta_0 + \beta_1 \times \mathbf{x} + \\ \sim \text{Normal}(0, \sigma^2)$$

where...

$\mathbf{y}$ = vector of home range sizes of coyotes β_0 = intercept β_1 = effect diff. of HR size for urban coyotes $\mathbf{x}$ = indicator of HR in urban (1) or not in urban (0) σ^2 = uncertainty / unknown variability

Statistical Model/Hypothesis

Example:

$$\mathbf{y} = \beta_0 + \beta_1 \times \mathbf{x} + \\ \sim \text{Normal}(0, \sigma^2)$$

Evidence of hypothesis support

β_1 is negative and **statistically clearly different**¹ than zero

Prediction

The expected outcome from a hypothesis. If agrees with data, it would support the hypothesis or at least not reject it.

1. Okay: Coyote home ranges are smaller in urban areas compared to non-urban areas
2. Better: Coyote home ranges in urban areas with high available food resources are smaller than coyote home ranges in urban areas with low available food resources and smaller than coyotes living in non-urban areas

Types of Studies

- Descriptive/Naturalist (not hypothetico-deductive)
- Hypothetico-Deductive Observational
- Hypothetico-Deductive Experimental

Take-Aways?

1. Study Objectives, Hypotheses, and Predictions

Big Data Problems

“The hidden Biases of Big Data” by Kate Crawford (2013)

“with enough data, the numbers speak for themselves”- Wired Magazine Editor

0,33	202,45	4862,46	1,21	1,45	0,85	8,51	6,35	300,48	1,15	3,38	0,30	2,19	0,78	1357,77	0,83	169,07	1,26	0,48	7,61	14,17
0,84	846,55	3484,47	1,64	3,52	0,52	6,76	9,41	1445,78	0,17	3,18	0,26	6,21	0,47	140,43	0,45	509,51	3,11	0,48	8,29	13,26
0,04	676,31	1427,13	2,87	3,28	0,62	1,97	9,46	1929,65	2,86	0,49	0,44	6,27	0,09	825,70	0,74	849,80	1,09	0,46	1,13	10,81
0,96	43,69	1611,10	0,77	3,69	0,26	6,47	3,11	893,24	0,58	1,07	0,61	4,03	0,65	1108,70	0,08	692,38	1,53	0,49	0,09	5,43
0,00	468,37	1310,22	2,71	0,70	0,47	6,93	9,02	1729,76	3,15	0,13	0,96	1,27	0,89	1101,45	0,49	198,52	0,20	0,73	4,08	5,49
0,71	762,70	4356,19	0,85	2,96	0,33	0,73	8,46	2463,40	2,51	1,43	0,95	6,26	0,83	1197,00	0,63	1338,64	3,22	0,45	4,91	14,47
0,95	84,85	457,84	2,98	3,55	0,29	4,14	6,52	416,30	2,30	2,07	0,62	6,94	0,76	675,53	0,45	1172,56	0,34	0,19	8,28	3,40
0,86	292,41	3760,23	0,77	0,59	0,25	4,86	0,06	999,49	1,79	2,81	0,63	6,39	0,64	1041,13	0,94	640,03	2,99	0,68	3,64	4,16
0,35	822,09	910,04	1,35	1,87	0,64	7,83	1,02	653,83	1,50	4,00	0,21	5,11	0,64	880,67	0,92	319,24	2,42	0,86	6,15	13,39
0,32	1389,45	4532,60	1,16	2,04	0,31	8,12	13,82	2281,72	2,02	3,17	0,45	3,48	0,89	1492,56	0,91	808,82	2,13	0,88	7,44	14,87
0,20	92,57	1702,20	3,71	1,66	0,04	0,96	13,73	2695,60	3,95	0,04	0,94	9,50	0,69	130,65	0,34	1200,34	1,39	0,64	8,65	9,02
0,12	794,10	4055,57	3,31	3,90	0,60	8,27	1,11	2765,63	3,85	3,52	0,48	1,10	0,07	960,19	0,17	1022,54	0,26	0,27	0,83	3,16
0,26	515,07	4431,19	3,36	0,10	0,44	7,40	8,96	581,78	0,59	3,04	0,12	9,75	0,89	675,84	0,14	106,58	3,39	0,47	6,61	5,54
0,30	181,71	4448,61	3,71	0,87	0,14	6,89	14,14	240,59	3,74	0,27	0,59	5,33	0,22	469,72	0,28	131,10	2,60	0,34	7,00	8,19
0,67	1021,48	559,76	1,88	1,42	0,28	5,23	0,83	890,83	1,05	1,87	0,21	8,26	0,38	1056,64	0,43	139,13	0,70	0,23	3,10	13,44
0,25	875,30	2191,24	3,22	3,10	0,10	9,60	2,21	1446,21	2,09	0,89	0,76	8,62	0,87	375,25	0,95	953,54	0,65	0,50	2,68	2,09
0,32	537,19	3976,58	2,19	0,88	0,99	3,28	1,79	2226,90	2,19	1,22	0,13	8,65	0,96	79,53	0,08	540,44	1,12	0,72	1,66	11,85
0,75	78,39	5775,90	1,76	2,19	0,40	8,39	14,70	497,83	2,88	1,62	0,58	8,71	0,87	230,22	0,39	1094,07	3,10	0,40	1,29	8,73
0,09	1220,90	1320,91	0,01	0,88	0,10	5,21	7,38	1757,83	0,57	0,57	0,13	5,40	0,62	1460,17	0,18	310,98	1,82	0,63	9,16	11,20
0,49	938,05	252,56	2,09	1,83	0,47	0,96	8,67	1892,22	3,64	2,13	0,93	7,78	0,73	861,45	0,11	562,53	1,69	0,67	1,71	0,49
0,05	1079,24	4665,57	0,18	0,30	0,34	7,75	8,75	1758,69	3,13	2,99	0,76	3,66	0,84	656,11	0,98	1267,72	3,61	0,40	0,01	6,05
0,84	1333,38	4464,09	0,14	2,81	0,21	0,94	0,99	1734,68	0,57	1,23	0,65	9,99	0,36	922,84	0,35	640,88	1,62	0,08	9,62	2,66
0,01	651,62	4129,64	2,23	1,94	0,02	5,38	8,18	348,67	3,64	1,23	0,15	7,13	0,77	140,68	0,73	342,64	2,66	0,04	8,82	13,42
0,82	1204,43	5247,37	1,44	2,70	0,20	7,50	0,55	2087,03	3,56	3,59	0,79	4,62	0,23	361,32	0,59	1418,51	3,75	0,17	5,84	2,08
0,74	1354,16	1633,83	1,56	1,07	0,82	3,90	10,01	307,46	3,09	3,33	0,52	7,11	0,54	478,56	0,96	438,50	3,50	0,03	1,29	5,35
0,13	741,44	485,14	1,25	0,84	0,84	7,59	0,30	271,90	3,98	2,94	0,27	8,50	0,66	123,32	0,96	462,81	1,02	0,93	6,54	0,49
0,42	1072,16	4361,62	0,12	1,44	0,26	7,68	9,73	228,51	0,85	2,01	0,26	1,76	0,40	1375,86	0,69	461,35	0,48	0,57	0,30	9,27
0,04	1385,13	2222,11	0,20	0,36	0,06	1,55	1,59	2402,97	0,52	3,31	0,21	6,95	0,29	325,73	0,17	814,54	0,27	0,18	0,24	7,78
0,44	1234,49	5302,87	3,09	1,51	0,76	6,82	7,28	530,72	1,78	1,57	0,57	4,13	0,15	126,16	0,33	460,85	1,62	0,07	2,04	9,10
0,94	1011,60	243,58	1,82	0,56	0,64	8,03	12,96	819,19	1,09	3,70	0,68	6,64	0,74	1350,17	0,54	830,75	2,45	0,31	2,73	6,43
0,76	137,92	4465,07	2,16	2,40	0,60	1,73	11,90	887,07	1,28	2,47	0,29	2,79	0,66	177,63	0,88	1339,96	1,85	0,00	5,16	9,69
0,47	23,55	1498,98	2,88	0,39	0,16	5,57	11,38	390,01	1,58	2,90	0,78	3,87	0,71	1334,75	0,51	408,46	3,02	0,21	7,18	11,83
0,19	848,16	4611,45	2,82	2,89	0,33	8,28	4,65	2046,08	1,64	0,82	0,91	7,42	0,93	1298,08	0,65	355,08	0,18	0,44	4,56	11,40
0,74	1089,34	5220,37	0,14	1,69	0,51	8,99	5,03	2443,55	0,84	1,18	0,53	7,29	0,30	1324,57	0,38	86,08	2,10	0,83	7,54	0,37
0,37	1407,80	3176,49	3,81	2,47	0,36	1,83	3,89	2491,19	2,13	3,83	0,00	6,97	0,82	1064,45	0,80	989,09	1,66	0,86	7,65	10,32
0,65	693,17	3245,95	3,05	1,86	0,40	2,56	8,90	352,00	3,25	0,60	0,10	2,68	0,13	1300,58	0,57	277,47	2,99	0,70	5,83	9,82
0,36	538,38	3348,08	1,25	0,17	0,85	5,89	1,36	1139,67	1,29	1,72	0,36	4,04	0,60	1102,77	0,47	599,78	2,01	0,82	1,05	4,71
0,19	1462,37	4621,41	1,66	2,26	0,64	0,07	2,19	2428,62	1,25	1,35	0,15	7,45	0,55	801,73	0,56	789,26	0,16	0,87	4,62	11,12
0,59	1052,61	1426,54	0,46	3,39	0,20	5,04	1,67	921,29	2,14	2,65	0,99	5,42	0,94	837,13	0,42	1149,36	3,20	0,08	5,03	3,39
0,25	680,56	5268,96	0,04	3,90	0,80	6,04	14,78	253,83	1,82	2,96	0,55	5,47	0,92	1181,12	0,43	319,59	0,55	0,41	8,73	5,36
0,40	627,42	5719,07	0,32	2,41	0,64	2,94	13,76	77,46	2,50	2,11	0,64	4,52	0,59	1445,81	0,57	1141,31	3,50	0,14	5,54	8,03
0,84	1328,00	5822,49	1,07	1,46	0,79	4,03	0,66	1282,37	3,60	3,58	0,33	4,88	0,88	941,87	0,46	955,18	2,48	0,58	3,18	7,20
0,15	180,07	3312,23	3,30	3,63	0,79	0,65	10,16	264,08	3,32	2,00	0,34	9,79	0,98	212,20	0,12	1439,36	0,97	0,34	5,69	8,44
0,52	253,05	5288,15	0,47	3,35	0,05	8,28	7,81	141,28	0,39	2,68	0,54	6,93	0,49	1355,70	0,48	1416,98	1,66	0,02	3,06	1,67
0,43	482,62	2222,80	0,10	0,75	0,65	7,42	8,76	879,10	3,49	1,55	0,25	8,52	0,62	1449,49	0,95	527,27	2,75	0,19	2,00	11,06
0,26	777,78	1632,70	2,88	1,89	0,56	7,35	8,57	716,36	3,39	1,27	0,44	2,70	0,81	549,92	0,88	457,23	3,70	0,96	7,01	6,87
0,90	1052,10	3363,54	1,55	1,38	0,39	1,18	9,84	2633,26	1,08	1,24	0,66	2,21	0,21	454,82	0,33	1475,64	1,60	0,64	8,21	2,49
0,67	894,53	147,80	1,48	2,40	0,03	7,09	7,72	1356,17	1,29	0,24	0,04	7,06	0,63	62,90	0,05	1185,04	3,52	0,59	0,46	1,15

Using assumptions to move from correlation

Data never speak by themselves. To derive meaningful conclusions on well-defined hypotheses, statistical models grounded in testable and untestable assumptions^{[57](#),[58](#),[59](#)}. The importance of statistical models, and statistical assumptions is well known.

- Correia et al. 2026. Best practices for moving from correlation to causation in ecological research. Nature Communications.

Big Data Problems

“The hidden Biases of Big Data” by Kate Crawford (2013)

...

Big Data Problems

The Annals of Applied Statistics (2018); Xiao Li Meng,

...

...

Sampling Processes == Human Design

Big Data Problems

![[Bradley et al. 2021 (Nature)]]

Big Data Problems

Using eBird data w/o accounting for sampling biases.

...

PNAS

Global abundance estimates for 9,700 bird species

Corey T. Callaghan  , Shinichi Nakagawa , and William K. Cornwell  [Authors Info & Affiliations](#)

May 17, 2021 | 118 (21) e2023170118 | <https://doi.org/10.1073/pnas.2023170118>

Extreme uncertainty and unquantifiable bias do not inform population sizes

Orin J. Robinson  , Jacob B. Socolar , Erica F. Stuber ,  +30, and Alison Johnston  [Authors Info & Affiliations](#)

March 1, 2022 | 119 (10) e2113862119 | <https://doi.org/10.1073/pnas.2113862119>

[Link1](#). [Link2](#).

The Questioning Scientist

In regard to *data* and *statistical models*, 21st century scientists should be pragmatic, excited, and questioning.

- How and why did these data come to be?
 - *understand how the data came to be*
 - *ask this even when you design the study, after data collection*

The Questioning Scientist

In regard to *data* and *statistical models*, 21st century scientists should be pragmatic, excited, and questioning.

- What do these data look like?
 - *visualize the data in many dimensions*
 - *keep in mind - not all outcomes are visible in data - example?*

The Questioning Scientist

In regard to *data* and *statistical models*, 21st century scientists should be pragmatic, excited, and questioning.

- How does this statistical model work?
 - *statistical notation, explicit and implicit assumptions, optimization*

The Questioning Scientist

In regard to *data* and *statistical models*, 21st century scientists should be pragmatic, excited, and questioning.

- How does this statistical model fail in theory and in practice?
 - *statistical robustness and identifiability*

Data vs. Information

. . .

- the question being asked of the data
- how the data came to be

The question and the data

Surveillance monitoring data will generally have *lower quality information* to answer post-hoc hypotheses when compared to a designed study with a priori hypotheses.



The goal of the question

- learn about the data (data summary)
- apply learning outside of the data to a ‘population’ (inference)
- learn about conditions relevant to but not observed in the data (prediction)

...

...

Take-Aways

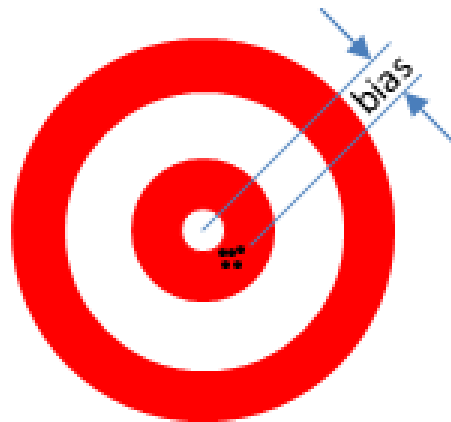
1. Study Objectives, Hypotheses, and Predictions
2. Big Data and Sampling

Which is worse?

- unbiased imprecise result
- precise biased result



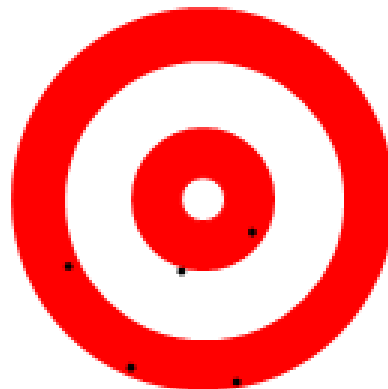
unbiased, precise



biased, precise



unbiased, imprecise



biased, imprecise

Inference and Prediction

Explanatory modeling focuses on minimizing (statistical) bias to obtain the most accurate representation of the underlying theory.

...

Predictive modeling focuses on minimizing both bias and estimation variance; this may sacrifice theoretical accuracy for improved empirical precision.

Inference and Prediction

...

BUT ...

Inference and Prediction

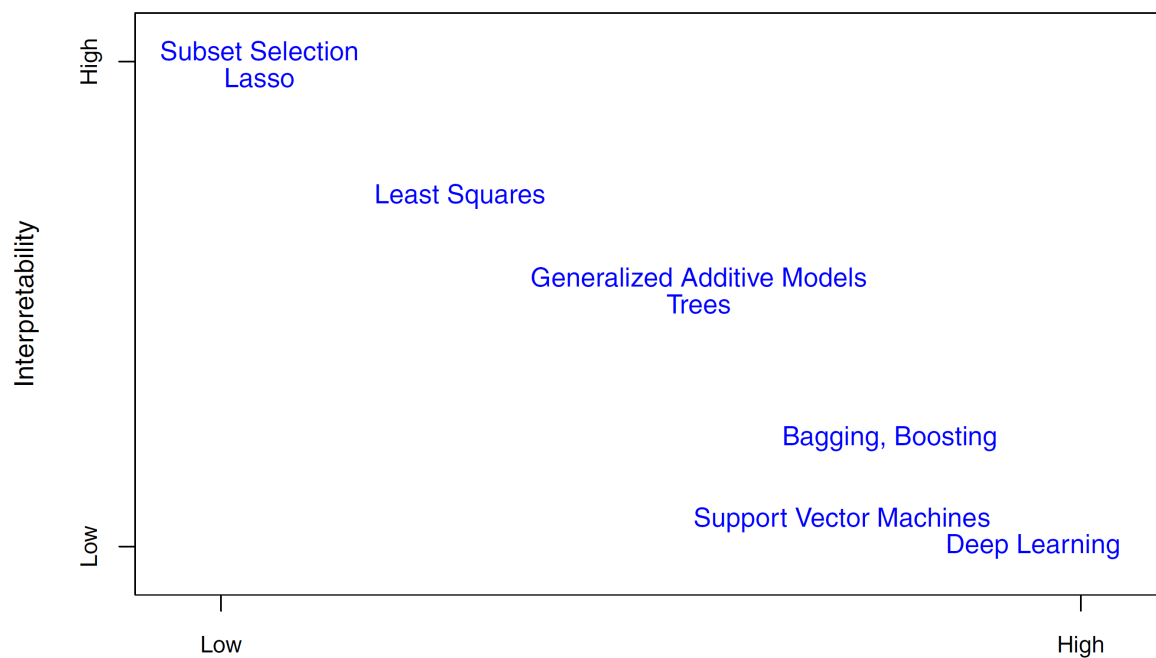


FIGURE 2.7. A representation of the tradeoff between flexibility and interpretability, using different statistical learning methods. In general, as the flexibility of a method increases, its interpretability decreases.

Take-Aways

1. Study Objectives, Hypotheses, and Predictions
2. Big Data and Sampling
3. Inference and Prediction

Design-Based

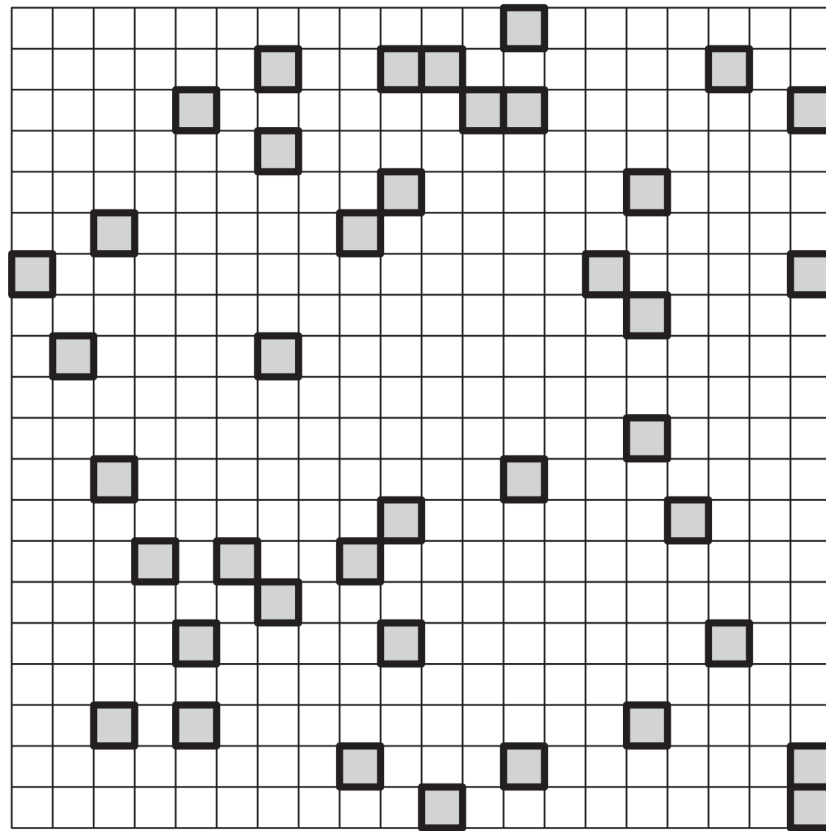


Figure 2.2. Another simple random sample of 40 units.

[Thompson, 2012. Sampling.](#)

The sample and population are what??

Design-Based

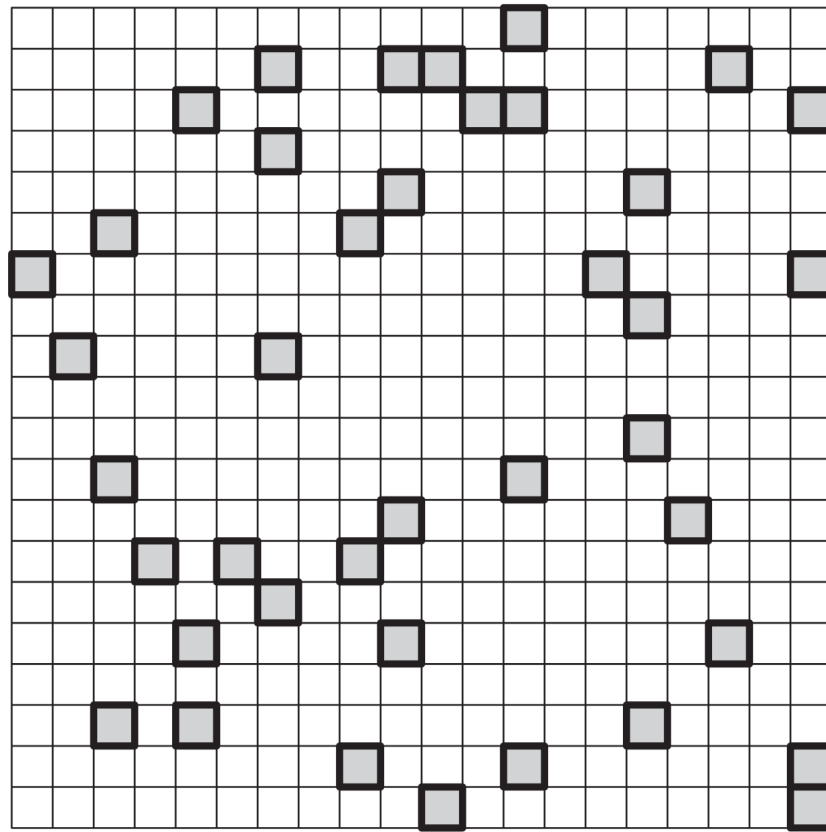


Figure 2.2. Another simple random sample of 40 units.

- inference relies on probabilistic assigning some units to be in the sample (e.g., random sampling).

Design-Based

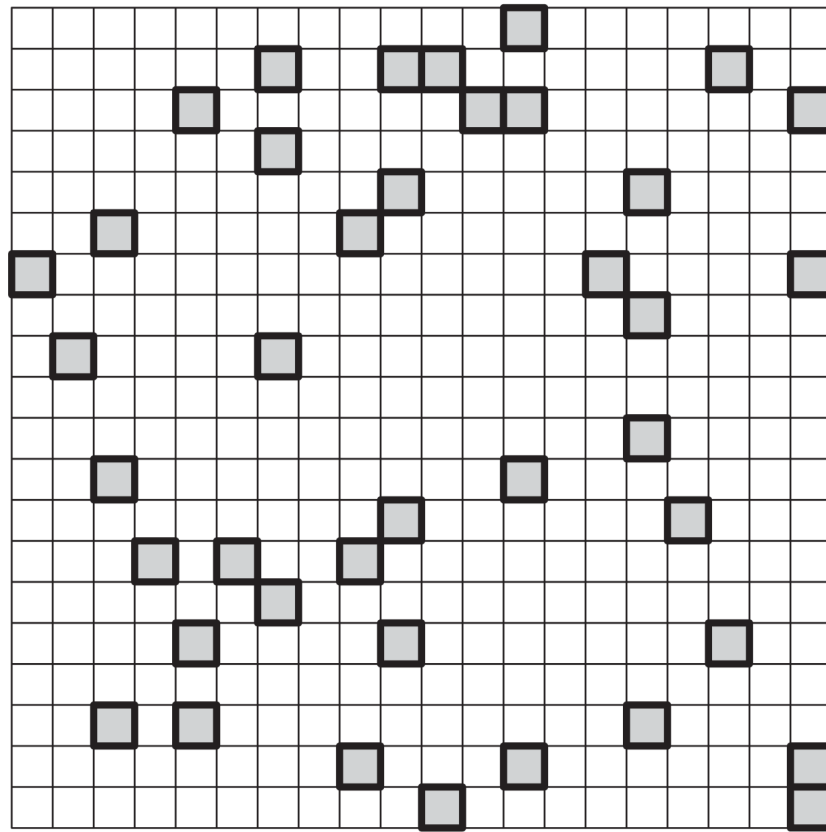
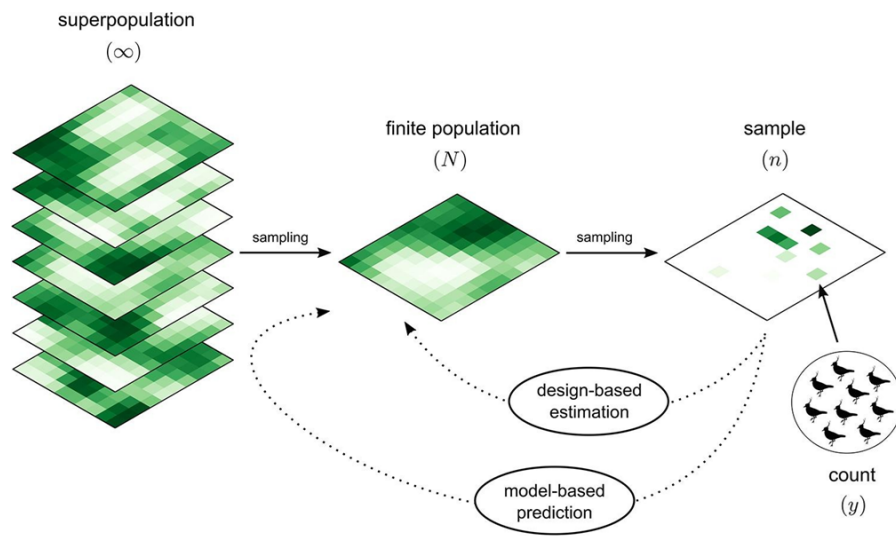


Figure 2.2. Another simple random sample of 40 units.

- the values themselves are held to be **fixed**, whereas the sampling process is random.

Two types of Inference



Aubry, P., & Francesiaz, C. (2022). On comparing design-based estimation versus model-based prediction to assess the abundance of biological populations. *Ecological Indicators*, 144, 109394.

Design-Based

- Key Strengths: the population of interest is often defined (e.g., grid area); does not relying on stochastic models representing the structure of the data for reliable inference

...

- Key Weaknesses: limited in application; still requires models to accommodate observational processes, such as detection probability

Design-Based

- $\mathbf{Y} = [y_1, \dots, y_N]$

...

- $\mathbf{y} = [y_1, \dots, y_n]$

Design-Based

- $\mathbf{Y} = [y_1, \dots, y_N]$
- The population mean is $\bar{Y} = \sum_{i=1}^N Y_i / N$ and the sample mean is $\hat{\bar{y}} = \sum_{i=1}^n y_i / n$

...

- The population mean describes?

...

- $\mathbf{y}$ is a random vector that has n random values, e.g., one sample of 4 cells.

...

$$\mathbf{y} = [y_1 \quad y_2 \quad y_3 \quad y_4]$$

...

$$\mathbf{y}' = \mathbf{y}^T = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Random Variable

[Wikipedia](#): A random variable (also called ‘random quantity’ or ‘stochastic variable’) is a mathematical formalization of a quantity or object which depends on random events.

...

We observe samples from the domain or population or sampling frame.

...

Samples are observed with some probability.

Statistic

- $\hat{\bar{y}}$ is a ‘statistic’ (# computed from a sample) and is also a random variable

...

- statistics have a sampling distribution, describing the probability associated to observing different values of the statistic

Design-Based Code

TRUTH

```
#random discrete uniform sampler
rdu <- function(n,lower,upper){
  sample(lower:upper,
         n,
         replace=TRUE
        )
}

N = 25

mat = matrix(rdu(N,
                 lower = 0,
                 upper = 400
                ),
             nrow=5, ncol=5
            )

mat
```

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	276	192	308	68	296
[2,]	13	329	247	261	2
[3,]	80	207	302	57	324
[4,]	124	294	315	298	260
[5,]	241	221	133	330	305

Design-Based Code

Random Sampling

```
n = 10
y = sample(
  c(mat),
  n,
  replace = TRUE
)

y
```

```
[1] 221 207 330 13 276 68 294 302 13 315
```

Estimator for the population mean

```
mean(y)
```

```
[1] 203.9
```

Design-Based Code

Sampling Distribution

```
# How many ways can we uniquely sample 10 things from 25
combs = function(n, x) {
  factorial(n) / factorial(n-x) / factorial(x)
}

combs(N, n)
```

```
[1] 3268760
```

Design-Based Code

Get every combination and then calculate the mean for each sample of 10

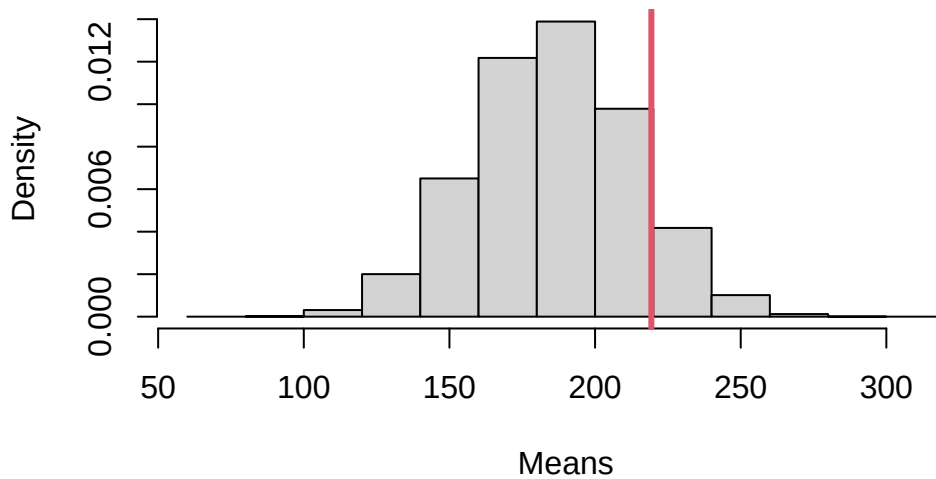
```
set.seed(5435)
all.combs = utils::combn(c(mat),
                          n
                        )
dim(all.combs)
```

```
[1]      10 3268760
```

```
mean.all.combs = apply(all.combs,
                        2,
                        mean
                      )
```

Design-Based Code

Sampling Distributioun of the Mean



Design-Based Code

OR, we can sample enough times to approximate it

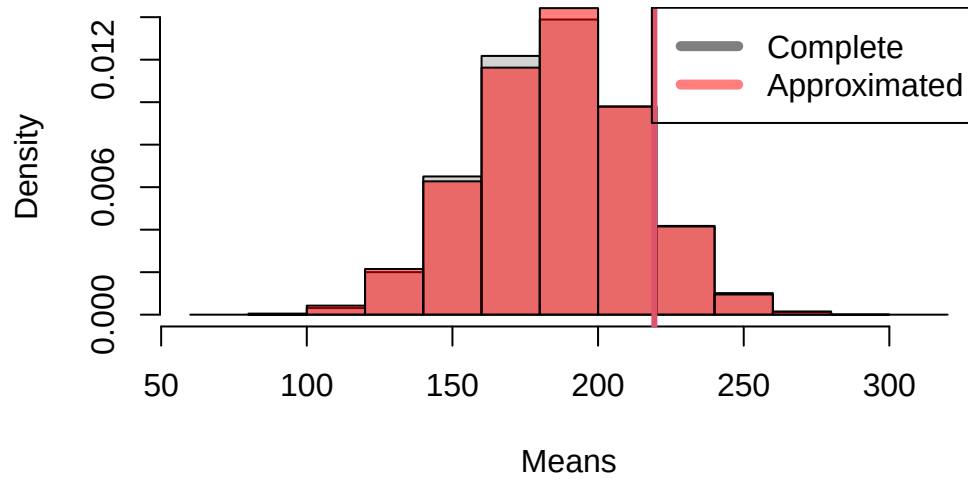
```
set.seed(54352)
sim.sampling.dist=replicate(2000,
                           sample(c(mat),n)
                           )
dim(sim.sampling.dist)
```

```
[1] 10 2000
```

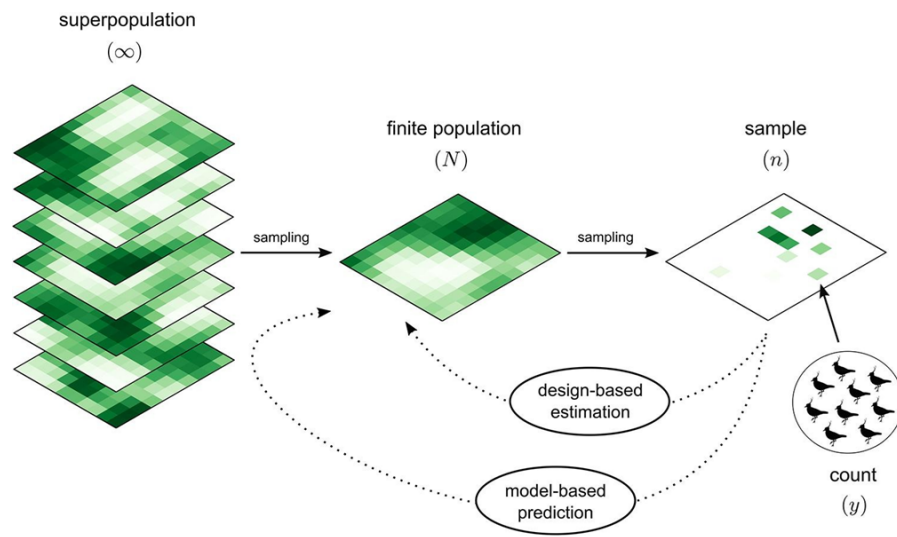
```
mean.samples = apply(sim.sampling.dist,2,mean)
```

Design-Based Code

Sampling Distributiun of the Mean



Two types of Inference



Aubry, P., & Francesiaz, C. (2022). On comparing design-based estimation versus model-based prediction to assess the abundance of biological populations. *Ecological Indicators*, 144, 109394.

Model-Based

Inference relies on ...

“a statistical model describing how observations on population units are thought to have been generated from a super-population with potentially infinitely many observations for each unit;” [Williams and Brown, 2019](#)

...

“The analysis need not account for sampling randomization, because the sample is considered fixed. However, the unit values are considered random.” [Williams and Brown, 2019](#)

Model-Based

BUT....

when linking ‘unit values’ in a model, we need to account for their dependence.

...

Randomization allows us to make conditional independence claims among data in our sample, thus the model is simpler.

...

$$P(y_2|y_1) = P(y_2)$$

Model-Based

- Key Strengths: Very flexible. Modeling is *magic*.

...

- Key Weaknesses:
 - Can be difficult to assess assumptions
 - Sampling frame is not always clear and thus the population you are inferring to is not entirely clear

Model-Based

[Wikipedia link for Poisson](#)

- $\mathbf{y} \sim \text{Poisson}(\lambda)$
- $y_i \sim \text{Poisson}(\lambda)$

...

- λ is the population mean and variance

...

- Population mean estimator $\lambda = \sum_{i=1}^N Y_i / N$
- Sample mean estimator $\hat{\lambda} = \sum_{i=1}^n y_i / n$

...

- Maximum-Likelihood Estimate (MLE)

Model-Based Code

```
# Create a function, to be replicated
lambda = 200

mat.fn = function(lambda){matrix(
  rpois(N, lambda=lambda),
  nrow=5, ncol=5
)}
```

Model-Based Code

```
# Repeat the function n.sim times
n.sim = 1000
list.mat = replicate(n.sim,
  mat.fn(lambda),
  simplify=FALSE
)

length(list.mat)
```

```
[1] 1000
```

```
...
```

```
# One realization from the super-population
list.mat[[1]]
```

```
      [,1] [,2] [,3] [,4] [,5]
[1,]  209  223  208  189  181
[2,]  222  175  194  187  190
[3,]  203  193  184  227  220
[4,]  199  210  203  212  198
[5,]  217  211  183  194  197
```

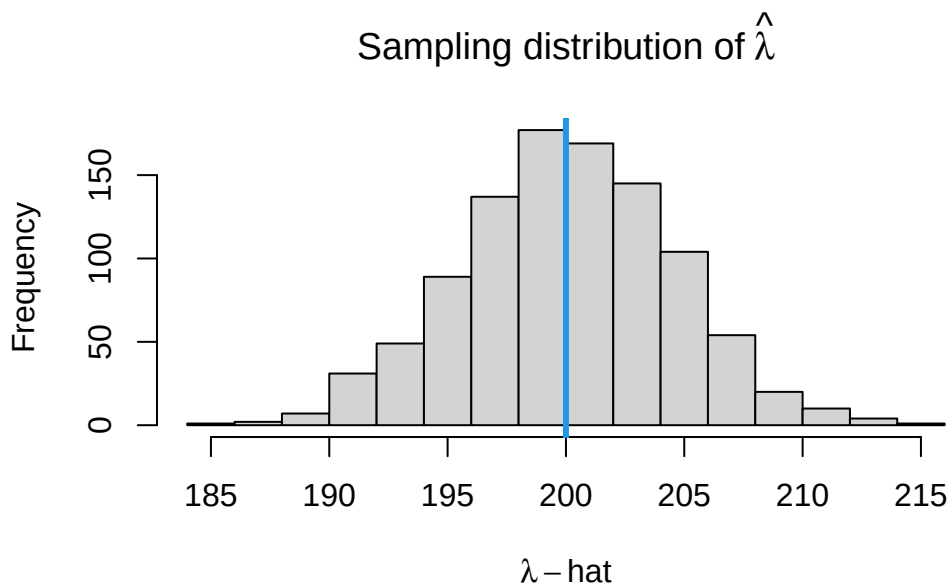
```
# Population mean for the first realization
mean(list.mat[[1]])
```

```
[1] 201.16
```

Model-Based Code

Sampling Distribution

```
samples.list = lapply(list.mat,FUN=function(x){sample(x,size=n)})  
lambda.hat = unlist(lapply(samples.list,FUN=mean))
```



Statistical Bias

the difference b/w the true value and the mean of the sampling distribution of all possible values; applies to design- and model-based sampling

Statistical Bias (Code)

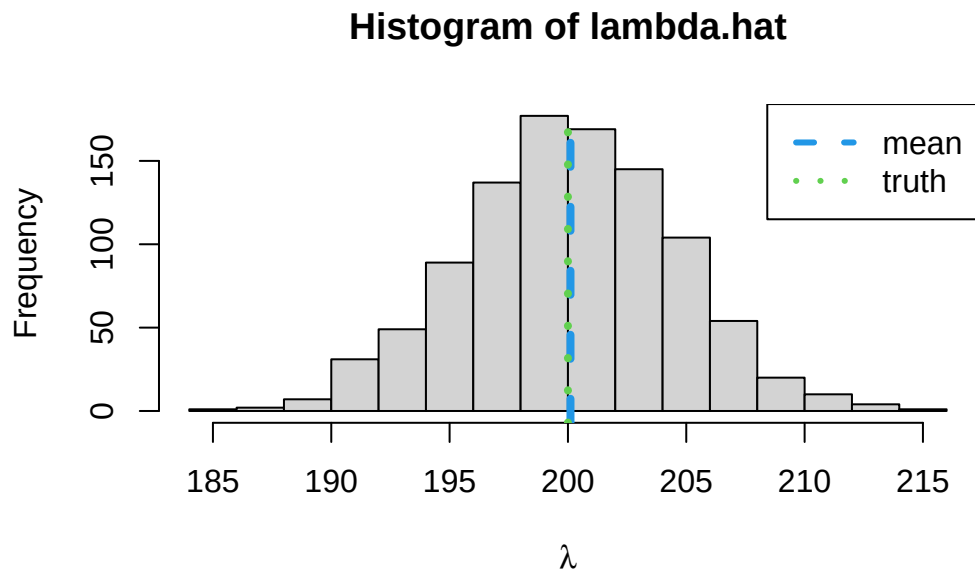
```
# Bias (in this case, Monte Carlo error)  
mean(lambda.hat) - lambda
```

```
[1] 0.1076
```



```
# relative bias
(mean(lambda.hat) - lambda)/lambda
```

```
[1] 0.000538
```



Precision of the mean (Code)

What is the probability that we will observe a mean within 5% of the truth?

...

We can calculate this using *Monte Carlo integration*

...

```
diff = 0.025*lambda
diff
```

```
[1] 5
```

```

lower = lambda-diff
upper = lambda+diff

index = which(lambda.hat>=lower & lambda.hat <= upper)

# Probability of getting a mean within 5% of the truth
length(index)/length(lambda.hat)

```

```
[1] 0.754
```

Take-Aways

1. Study Objectives, Hypotheses, and Predictions
2. Big Data and Sampling
3. Inference and Prediction
4. Model-Based vs Design-Based Inference

Lab

Objectives

- Introduce R Markdown
- Use simulation and design-based sampling to investigate bias and precision

Lab Setup

Let's add some more *reality* in our work while using design-based sampling in R.

...

Objective: Evaluate sample size trade-offs for estimating white-tailed deer abundance throughout Rhode Island.

...

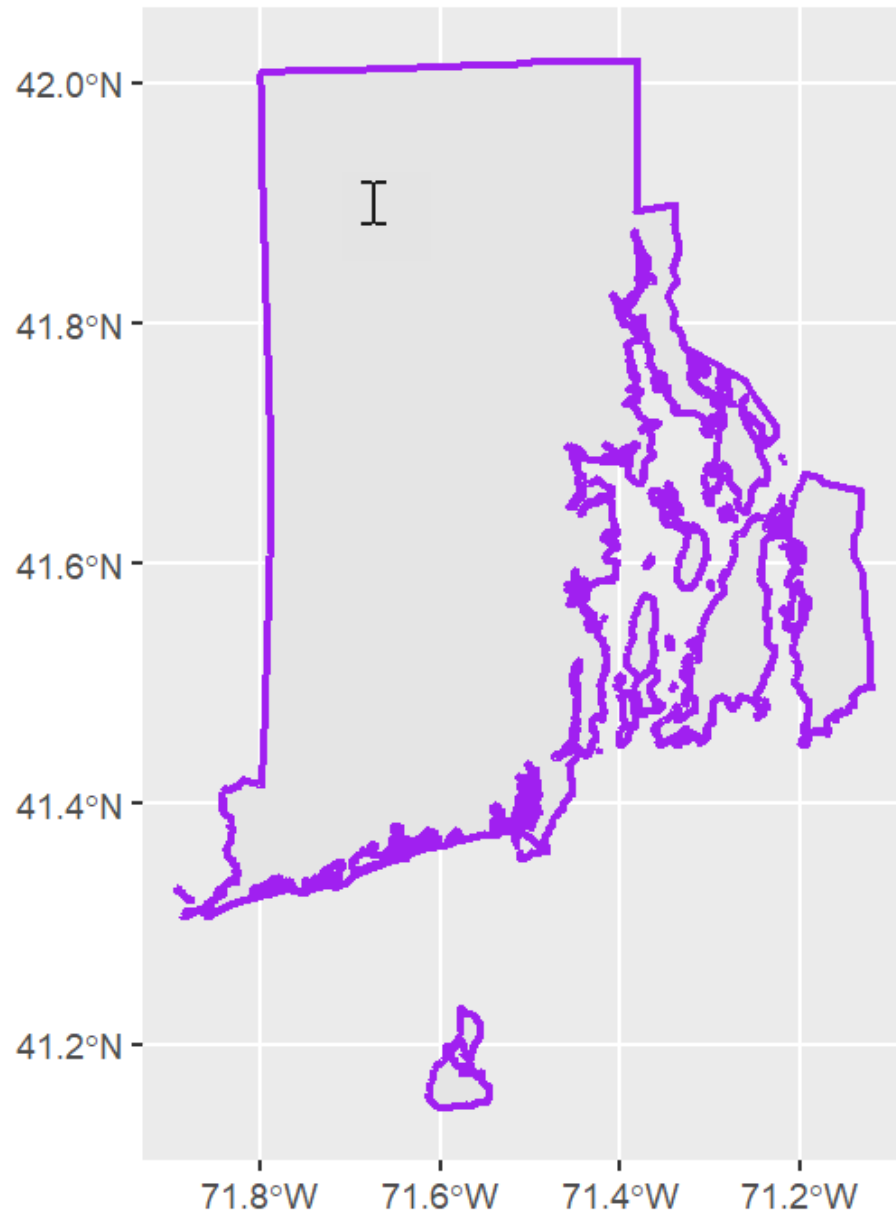
Methodology: Count deer in 1 sq. mile cells using FLIR technology attached to a helicopter.



Lab Setup

Steps to consider

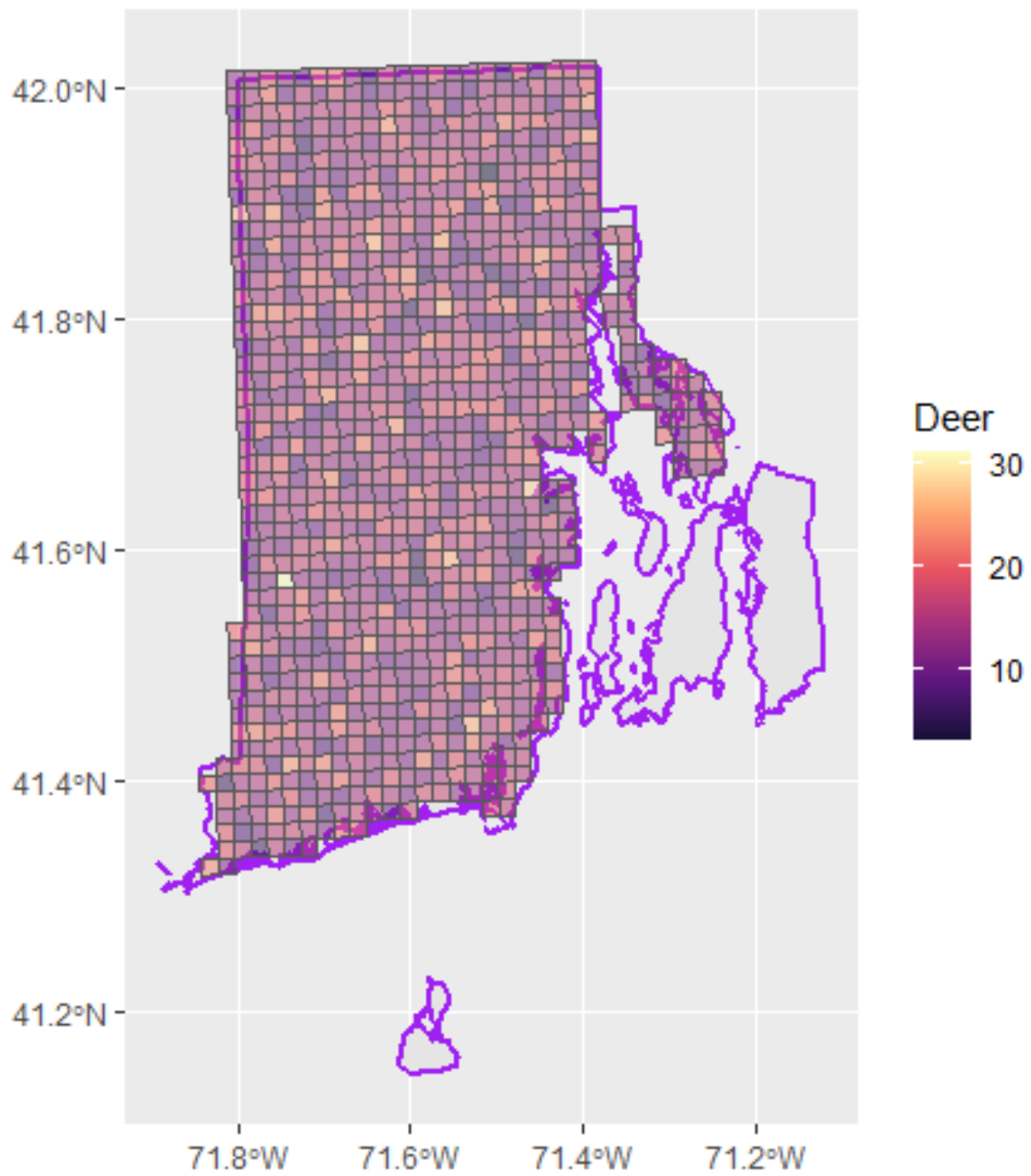
- **Sampling Frame**
 - all of RI or some subset



Lab Setup

Steps to consider

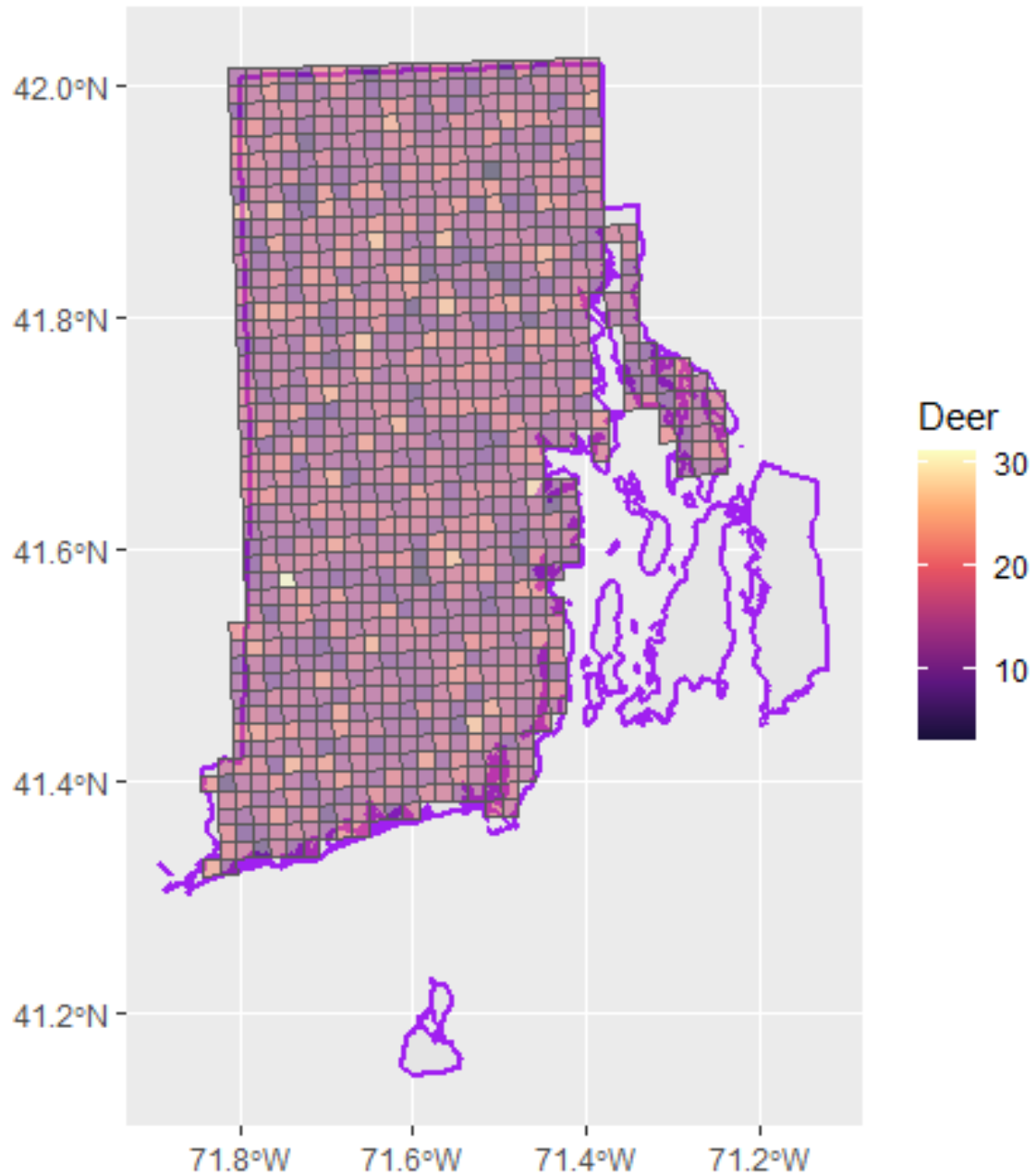
- “Truth”
 - how many deer per cell; how variable



Lab Setup

Steps to consider

- Sampling Process
 - how to pick each cell

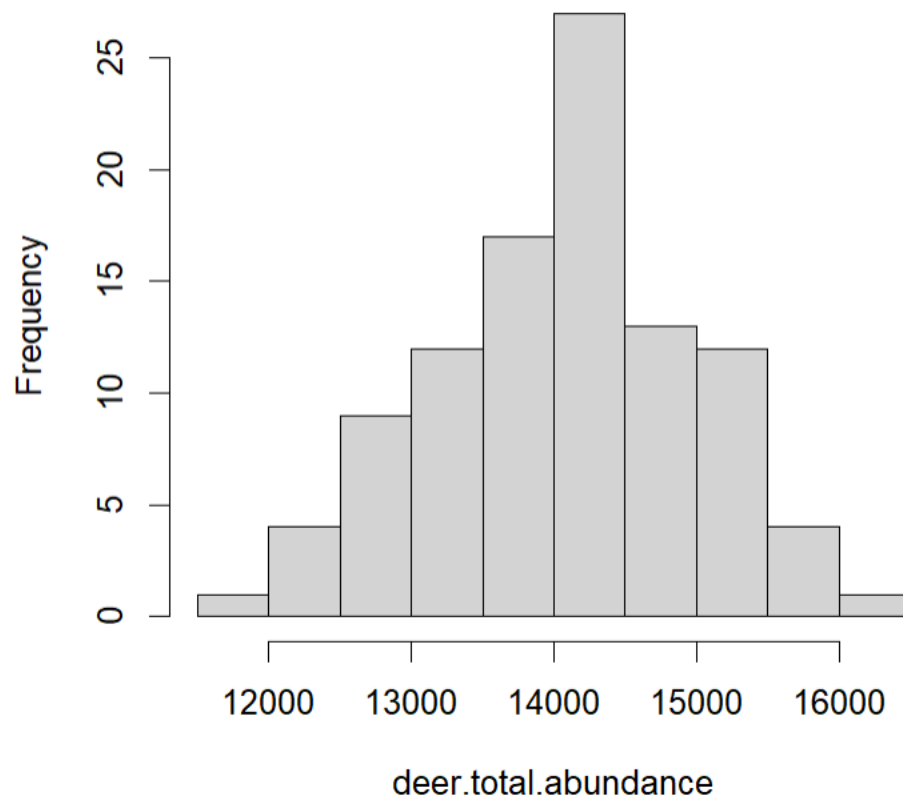


Lab Setup

Steps to consider

- **Estimation Process**
 - estimate total deer population from the sample
- **Criteria to Evaluate**
 - use sampling distribution of deer abundance estimate or some other statistic

Histogram of deer.total.abundance



Go to code